

Impact of Telecom Connectivity on Digital Economic Growth in India

A State-wise Data Science Analysis

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Abstract

Considering Bharti Airtel as a benchmark for telecom within the context, this paper explores the linkages between digital economic participation and telecom connectivity in 36 states in India. By conducting a careful process of data cleaning, standardization of states, and feature engineering, a consistent dataset was compiled, which consisted of telecom subscribers per state, percentage 5G coverage, teledensity, literacy, Gross State Domestic Product (GSDP), and electronic transactions. The index was normalized based on the Min-Max normalization technique, whereby each of the four components received an equal weightage to compute the Digital Connectivity Index (DCI). K-Means Clustering and econometric interaction analysis with HC3 robust standard errors have been performed along with seven different predictive models, including three types of linear regressions.

The literacy variable significantly performed better than the telecom infrastructure variables in predicting the digital economy as the primary variable for all factors considered. Overfitting occurred due to the small number of states used as training samples by the machine learning algorithms. The good cross-validation performance levels of the algorithms (Random Forests CV R^2 : 0.472, XGBoost CV R^2 : 0.649), however, did not affect the results. Telecom coefficients proved difficult to interpret in the parametric model owing to high multicollinearity between the teledensity and 5G penetration variables ($VIF > 32,000$). This study concludes that whereas telecom infrastructure is a necessary but not sufficient driver of the digital economy, institutional readiness and human capital become important additional inputs. The Digitally Advanced, Emerging, and Underserved state archetypes are distinguished via strategic cluster segmentation.

Keywords: Telecom connectivity, digital economic growth, 5G penetration, literacy, machine learning, India, Elastic Net, Random Forest, XGBoost, K-Means clustering, DCI

1. Introduction

The development of mobile payment systems, digitalization initiatives driven by governments, and extensive use of cell phones are some of the elements that have played their part in the evolution of India's digital economy in the last decade. In particular, the introduction of UPI in 2016 has resulted in an unprecedented speed of embracing digital payment methods in developing countries. Still, despite being introduced in the same telecom environments, jurisdictions demonstrate distinct participation in digital activity. Therefore, it becomes evident that digital adoption requires more than just access to technology.

Bharti Airtel, being the second largest telecommunication operator in India based on number of customers, provides an essential strategic point of view for the assessment of infrastructure adoption dynamics. It is much more convenient to conduct research on whether higher infrastructure leads to proportional benefits from digital economy on the state level thanks to the rapid implementation of 5G technology in use by Airtel that is expected during 2024-2025.

The three key research questions posed in this study are as follows:

1. To what extent do electronic transactions per capita serve as measures of the state level engagement in digital economic activity and how does it relate to telecom infrastructure (teledensity, 5G usage, subscriptions increase)?
2. Does predictive validity enhance through application of the Digital Connectivity Index, a composite measure, in contrast to the isolated usage of individual infrastructure variables?
3. What type of interdependence exists between telecom infrastructure and literacy rates and is there an importance of human capital to the success of infrastructure investment?

The study contributes to the growing body of literature on digital divide and infrastructure-based growth in developing countries.

2. Data & Methodology

2.1 Dataset Construction

Six state level data sources, consisting of 36 states and union territories in India, were compiled for analysis:

- Subscriber and teledensity data from the Telecom Regulatory Authority of India (TRAI) (2024)
- Share of market and rollouts of Airtel 5G per state
- Data on national payments ecosystem: transactions through electronic means by state
- Population and literacy data based on census
- Gross State Domestic Product (GSDP) in ₹ crore (current price)
- Density and geographic area

To resolve any issues regarding inconsistent data sources, standardization of state names was done. As a precautionary measure suitable for our cross-sectional data sample, mean value imputation by columns was done for missing data (mostly for union territories). In total, there were 36 observations with 16 variables.

2.2 Feature Engineering

Five derived variables were constructed to better capture the theoretical constructs of interest:

- Engineered Teledensity: $\text{Number of total subscribers/million} / \text{Population} \times 100$, where a teledensity measure normalized on a per-population basis has been considered that differs from the raw teledensity data provided by TRAI.
- 5G Penetration: $5\text{G subscribers/population} \times 100$, representing the adoption of the latest technology in terms of advanced connectivity as a proportion of the state's population.
- communication-literacy Synergy: $\text{Teledensity} \times \text{Literacy rate}$, where the interaction captures the relationship in which communications infrastructure complements human capital.
- $\log(\text{GSDP})$: Log transform of GSDP normalizes the difference in scale between big and small states and also corrects extreme right skew in GSDP.
- DCI: Initially, the four components of the Digital Connectivity Index (DCI) were given equal weights of 0.25. After a rigorous review, this arbitrary weighting was replaced with a PCA-derived index (DCI_PCA), in which component loadings are decided by the data's variance structure rather than by hand. The DCI score is derived from the first principal component, which represents the dominant axis of variation across teledensity, 5G penetration, subscriber growth, and literacy. This method is more convincing as a latent connection metric and removes the theoretical arbitrariness of equal weighting. The PCA-derived DCI was shown to function

similarly to its component literacy measure alone as a diagnostic note, indicating that aggregation across telecom and human capital dimensions somewhat dilutes rather than compounds the signal

In order to adjust for distributional skew and heteroscedasticity, the transformed target variable, $\log(e_transactions_per_1000)$, is calculated as $\log(1+x)$.

The extreme multicollinearity observed between engineered teledensity and 5G penetration ($VIF > 32,000$) reflects a variable construction issue: both metrics share the same denominator (state population) and are highly correlated numerically across states. Following mentor review, regression specifications were simplified to avoid including both variables simultaneously. Subsequent models prioritise non-redundant telecom indicators, and the interaction term (telecom $\times$ literacy) is retained as the primary infrastructure-side variable given its theoretical motivation and lower collinearity.

2.3 Analytical Framework

The four-stage process involved in the analysis was: (1) Exploratory Data Analysis to investigate the distribution of variables and their relationship; (2) Prediction Modeling involving seven specifications; (3) State Clustering by means of K-Means clustering; and (4) Econometric analysis utilizing Ordinary Least Squares regression along with heteroskedasticity-robust standard errors HC3.

In case of hyperparameter tuning through GridSearchCV, all the predictive models adopted an 80-20 train-test split ratio with random state 42 and 5-fold cross-validation. This approach has been preferred for reasons of reliability because of the small sample size available in the dataset ($n=36$) rendering train test split highly volatile. Leave-One-Out Cross-Validation (LOOCV) was additionally implemented as the primary evaluation criterion for small-sample robustness, as it maximises training data at each fold and is statistically more appropriate than k-fold CV when $n = 36$. The 80/20 train-test split is retained for comparison and exploratory benchmarking only.

3. Exploratory Data Analysis

3.1 Regional Heterogeneity

The dataset exhibits substantial regional heterogeneity across all key variables. Electronic transactions per 1,000 population ranged from approximately 2,991 in the lowest cluster to 17,858 in West Bengal, the leading state. Table 1 summarises the top-performing states across three metrics.

Table 1: Top-Ranked States by Digital Activity, DCI, and Teledensity

Rank	Top Digital States	Top DCI States	Top Teledensity States
1	West Bengal (17,858)	Northeast (0.506)	Northeast (104.3)
2	Haryana (10,230)	Delhi (0.469)	Northeast (60.4)
3	Himachal Pradesh (8,892)	Northeast (0.399)	Northeast (50.6)
4	Tamil Nadu (8,237)	Punjab (0.379)	Northeast (33.6)
5	Northeast (7,939)	Northeast (0.375)	Northeast (24.5)

A notable feature of the data is the dominance of Northeast states in teledensity rankings. This reflects geographic concentration effects — smaller, geographically compact union territories and northeastern states register artificially elevated teledensity when subscribers from adjacent areas are attributed to them. This structural data artefact underscores the importance of the engineered teledensity variable, which normalises for population size.

3.2 Bivariate Relationships

Scatter plots with regression overlays for the four primary predictors against the target variable revealed consistently weak linear relationships. Literacy showed the strongest visual association with e-transactions, followed by GSDP. Teledensity and 5G penetration exhibited highly diffuse scatter with no discernible linear trend, suggesting that telecom infrastructure alone does not determine digital economic outcomes.

The correlation analysis confirmed moderate positive correlations between literacy and digital transactions, while telecom variables showed low and inconsistent correlations with the target. The DCI correlated moderately with e-transactions, reflecting its incorporation of literacy as a component. This pre-modelling evidence already points toward the central finding: human capital, not infrastructure density, is the primary driver of digital adoption.

3.3 Multicollinearity Structure

The correlation heatmap revealed severe multicollinearity among telecom variables. The engineered teledensity and 5G penetration measures are near-perfectly collinear ($VIF > 32,000$ for both, as confirmed by formal VIF analysis in Section 5). This collinearity structure is a fundamental constraint on the

interpretability of parametric models that include both variables simultaneously, motivating the use of regularisation and dimensionality reduction in subsequent modelling stages. Importantly, the extreme VIF values ($>32,000$) between teledensity and 5G penetration indicate a variable construction issue rather than just a statistical annoyance. This is because both metrics are calculated using population as the denominator, and the percentage of 5G subscribers in each state is almost constant, making them the same variable at different scales. Instead of only regularization, this calls for variable reformulation.

4. Predictive Modelling Results

4.1 Model Performance Overview

Table 2 depicts the overall performance of all seven models. Generalization capacity measured through CV R^2 (since there is only a small sample available) is the key metric to be considered for evaluation purposes. The elastic net, as well as all linear regression models, returned a negative value of the CV R^2 measure, meaning worse than baseline results regarding generalization capability. Both random forest and XGBoost performed the best under the cross-validation standard with positive CV R^2 measures, namely 0.472 and 0.649, correspondingly.

Table 2: Model Performance Comparison

Model	Test R^2	MAE	RMSE	CV R^2 (mean)
LR 1 — Economic Baseline	-2.014	0.571	0.716	-0.280
LR 2 — Telecom + Economic	-1.668	0.492	0.674	-0.468
LR 3 — DCI Composite	-2.288	0.564	0.748	-0.201
Elastic Net ($\alpha=0.1$, $L1=1.0$)	-1.138	0.480	0.603	-2.041
Random Forest	-1.887	0.514	0.701	0.472
XGBoost	-3.804	0.678	0.904	0.649

In tree algorithms, there is a marked contrast between the CV and test R^2 values, whereby while XGBoost has a CV R^2 of 0.649 and a test R^2 value of -3.804, the Random Forest algorithm has a CV R^2 of 0.472 and a test R^2 of -1.887. This trend highlights the inherent volatility of splitting data into 80-20 ratios on a sample size of 36 observations, and this is not indicative of model failure, but overfitting of the test dataset.

4.2 Linear Regression Models

Once standardized, Model 1 (Economic Baseline: GSDP + Literacy) produced a CV R^2 of -0.280, with literacy having a positive coefficient (0.269) and GSDP exhibiting a negative coefficient (-0.137). Given that high-GSDP regions have a history of having poorer digital transaction activity, compared to medium-GSDP regions with good payment systems and literacy rates, this indicates a dominance effect.

Incorporating the teledensity feature resulted in an increased test set fit but a reduction in RMSE (0.674 vs. 0.716) and CV R^2 (-0.468) for Model 2 (Telecom + Economic: log GSDP, Teledensity, Literacy), meaning that the latter had worse cross-validation performance. In either case, literacy was found to have the greatest positive coefficient (0.273).

Even though DCI theoretically makes sense as a broad measure of connectivity, it turned out to perform the poorest of all the linear models, yielding a CV R^2 of -0.201. This means that while a composite factor can be beneficial for classification, it reduces the resolution in the regression signal.

4.3 Elastic Net Regression

The Elastic Net model — tuned to $\alpha = 0.1$, L1 ratio = 1.0 (equivalent to Lasso) — effectively eliminated all telecom-specific coefficients, retaining only literacy (coefficient: 0.205) and log GSDP (coefficient: -0.025). This automatic feature selection result is analytically significant: when regularisation is applied to correct for multicollinearity, the model discards all telecom infrastructure variables in favour of the human capital proxy. This is among the strongest evidence in the analysis for the primacy of literacy over infrastructure in predicting digital adoption. It should be noted that the optimal hyperparameters ($\alpha = 0.1$, L1 ratio = 1.0) make this equivalent to a Lasso regression rather than a true Elastic Net, which would explore the L1/L2 space jointly. The L1 ratio = 1.0 result is itself informative — it suggests that no combination of L1 and L2 penalties outperforms pure variable selection in this data context.

4.4 Random Forest & XGBoost

Table 3 presents the feature importance rankings from both ensemble methods. Literacy accounts for 58.4% of the Random Forest's predictive contribution and 46.9% of XGBoost's — by far the largest share in both cases. The DCI and Telecom–Literacy interaction term rank second and third respectively in XGBoost, while log GSDP ranks second in Random Forest, suggesting that economic scale plays a modestly stronger role in the forest specification.

Table 3: Feature Importance Rankings — Ensemble Models and Elastic Net Coefficients

Feature	RF Importance	XGB Importance	EN Coefficient
Literacy	0.584	0.469	0.205
Digital Connectivity Index	0.105	0.188	0.000
Telecom–Literacy Interaction	0.079	0.159	0.000
log(GSDP)	0.086	0.023	-0.025
Subscriber Growth Rate	0.055	0.073	0.000
Teledensity	0.061	0.025	0.000
5G Penetration	0.031	0.062	0.000

The consistently high importance of the Telecom–Literacy interaction term (7.9% in RF, 15.9% in XGBoost) provides partial support for the complementarity hypothesis: telecom infrastructure matters, but its economic effect is contingent on the human capital base of the receiving state. Teledensity and 5G penetration individually contribute less than 7% importance in both models, further corroborating the Elastic Net finding.

These ensemble results should be read as exploratory feature ranking exercises rather than predictive performance claims. Simplified model architectures — reduced `max_depth` and `n_estimators` — were applied to limit overfitting on the small sample. The consistent ranking of literacy as the dominant feature across both models, despite their structural independence, constitutes the most reliable takeaway from the ensemble analysis.

5. State Segmentation — K-Means Clustering

To divide Indian states into different digital readiness profiles, K-Means clustering ($K = 3$, chosen via elbow technique) was applied to four standardized variables: teledensity, 5G penetration, DCI, and e-transactions per 1,000. The mean values for each cluster are shown in Table 4.

Table 4: K-Means Cluster Profiles ($n = 36$ states, $K = 3$)

Cluster	Teledensity	5G Penetration	DCI	E-Tx / 1,000
0 — Digitally Advanced	0.016	144,500	0.304	7,094
1 — Emerging	0.465	4,314.2	0.280	6,391
2 — Underserved	0.001	5.167	0.163	2,991

Cluster 0 (Digitally Advanced) captures states with moderate teledensity but high e-transaction volumes (mean: 7,094 per 1,000) and elevated 5G penetration. These states likely represent urban-dense or high-literacy markets where existing infrastructure is being actively utilised for digital commerce. Cluster 1 (Emerging) is characterised by high 5G penetration (4,314) relative to its DCI and e-transaction scores, suggesting infrastructure-ahead-of-adoption dynamics — precisely the scenario that the literacy-primacy finding predicts. Cluster 2 (Underserved) records the lowest values across all four dimensions and represents states where simultaneous investment in connectivity and digital literacy programmes is required.

When it comes to deploying infrastructure through the model, the clustering provided is operationally relevant as follows: Cluster 1 countries can receive literacy programs as well as supply side infrastructure; Cluster 2 countries require basic infrastructure investments before digitalization initiatives can be implemented; and Cluster 0 requires better 5G service quality as well as business applications.

6. Econometric Analysis

6.1 Multicollinearity — VIF Analysis

Variance Inflation Factor scores for all the features are presented in Table 5. The presence of very high multicollinearity ($VIF > 32,000$) among teledensity and 5G penetration is evident, confirming that these two features have nearly identical information that cannot be incorporated simultaneously in parametric

analyses without inducing high instabilities. Since they are part of the DCI construct, literacy and log GSDP have relatively high VIF values (259 and 144, correspondingly). Of all the telecom variables considered, only the subscriber growth rate shows a proper VIF score (4.1).

Table 5: Variance Inflation Factors — Full Feature Set

Feature	VIF
5G Penetration	32,462.5 — Severe
Teledensity	32,445.7 — Severe
Literacy	259.0 — High
log(GSDP)	143.6 — High
DCI	42.8 — Moderate-High
Subscriber Growth	4.1 — Acceptable

These VIF values indicate that teledensity and 5G penetration should not be treated as independent variables in any parametric model. The correct analytical response is variable reformulation — for instance, using only subscriber growth rate (VIF: 4.1) as the telecom predictor and the interaction term as the complementarity measure — rather than relying on regularisation to manage a multicollinearity problem that originates in variable construction. These VIF values indicate that teledensity and 5G penetration should not be treated as independent variables in any parametric model. The correct analytical response is variable reformulation — for instance, using only subscriber growth rate (VIF: 4.1) as the telecom predictor and the interaction term as the complementarity measure — rather than relying on regularisation to manage a multicollinearity problem that originates in variable construction.

6.2 Robust Econometric Model

In fact, all variables within the interaction specification were included in an ordinary least squares (OLS) regression with HC3 heteroskedasticity-robust standard errors, whose results are reported in Table 6. With R² equal to 0.302 and adjusted R² equal to 0.211, the entire model is significant ($F(2) = 3.980$, $p = 0.010$). With the sole exception of the constant ($p = 0.041$), none of the individual variables in the above specification are significant at the customary 5% level.

Table 6: HC3 Robust Regression — Interaction Model ($n = 36$)

Variable	Coeff.	Std. Err.	z-stat	p-value
Intercept	6.810	3.327	2.047	0.041*
Literacy	0.036	0.031	1.135	0.256
Teledensity	-115.104	598.109	-0.192	0.847
log(GSDP)	-0.063	0.142	-0.442	0.658
Telecom × Literacy	1.572	8.135	0.193	0.847

his finding is consistent with the highly multicollinear nature of the model, as highlighted in the above VIF analysis: while the model's explanatory power is preserved, the joint presence of collinear variables increases variance and makes any individual coefficients statistically uninformative. Notwithstanding, the signs of both the literacy coefficient (0.036) and the interaction coefficient (1.572)—the former being positive and the latter implying that the effect of literacy on digitalization is enhanced by the telecom sector—are consistent with theoretical expectations. However, neither coefficient achieves statistical significance at the state-level data set.

Autocorrelation in the residual series does not exist since the Durbin Watson statistic is 1.757. Multicollinearity exists in the model due to the high condition number of 6.19×10^1 , indicating that regularization or reduced model dimensions are needed.

The F-statistic degrees of freedom ($df_{\text{model}} = 4$) reflect the four predictors in the specification. The model's overall significance ($p = 0.010$) is maintained despite individual coefficient insignificance, consistent with the multicollinearity diagnosis: the predictors collectively explain variation that cannot be cleanly attributed to any single variable given their correlation structure.

7. Discussion

7.1 The Literacy Primacy Finding

The most robust finding across all seven models and both econometric specifications is the primacy of literacy in predicting digital economic activity. This result is consistent across linear regression coefficients, Elastic Net's automatic selection (which retained only literacy among all predictors), Random Forest feature importance (58.4%), and XGBoost feature importance (46.9%). Literacy acts as a proxy for the broader construct of digital readiness — encompassing not only the ability to read and navigate digital interfaces but also the underlying cognitive and institutional infrastructure that enables digital commerce participation.

This finding challenges a purely infrastructure-led view of digital development and aligns with the literature on digital complementarities (Brynjolfsson & McAfee, 2014; World Bank, 2016). Physical connectivity is a necessary but not sufficient condition: without the human capital to translate connectivity into economic activity, infrastructure investment yields limited returns in digital transaction outcomes.

7.2 Telecom Infrastructure — Necessary but Not Sufficient

The consistent near-zero or negative contribution of teledensity and 5G penetration to predictive performance across linear and regularised models does not imply that telecom infrastructure is irrelevant. Rather, the evidence suggests that at the current stage of India's digital development, the variance in digital economic outcomes across states is explained primarily by demand-side factors (literacy, digital readiness) rather than supply-side infrastructure availability. States with high teledensity but low literacy — as exemplified by Cluster 1's infrastructure-ahead-of-adoption profile — confirm this interpretation.

The positive and moderately large interaction coefficient between teledensity and literacy (1.572 in the robust model, 15.9% importance in XGBoost) provides partial support for a contingency view of infrastructure: connectivity investments generate stronger digital economic returns in states with higher human capital endowments. This suggests a sequenced policy logic — foundational literacy and digital education investments should accompany or precede infrastructure deployment in underserved regions.

7.3 Limitations

A number of limitations should be noted. Firstly, the generalization instability of trees based methods with regards to the test sample represents a limitation rather than an error in the model specification because the sample size ($n = 36$) is inherently small for machine learning approaches. Secondly, correlations can be distorted by additional noise resulting from using the mean value imputation technique for handling missing data in Union territories. Thirdly, there exists a structural deviation in the aggregate measure of the tele-density indicator for the group of Northern states that can lead to a higher measurement error. Lastly, as a consequence of the cross-sectional nature of the data used, it is not possible to make any causal conclusions, meaning that the relationship between adoption of new technologies and literacy might actually be driven by some unobserved factors.

Beyond simple notation, a number of constraints need to be explicitly acknowledged. First, this analysis treats ML outputs as exploratory feature rankings rather than predictive claims; however, future work should use either district-level data (700+ observations would legitimately support these methods) or panel data across multiple years to establish both sample adequacy and temporal identification. This is because the sample size of $n = 36$ is fundamentally incompatible with the machine learning toolkit deployed. Second, in subsequent iterations, variable reformulation should be used to address the severe multicollinearity amongst engineered telecom variables, which is a construction artifact. Third, sensitivity studies that exclude these observations are an essential robustness step because mean imputation for union territory observations creates significant fake data points.

8. Conclusions

This study examined seven predictive models and two econometric specifications to assess the relationship between telecom connectivity and digital economic growth across 36 Indian states. The central conclusion is that:

$$\textit{Digital Economic Growth} = \textit{Infrastructure} \times \textit{Human Capital} \times \textit{Institutional Readiness}$$

Telecom infrastructure — teledensity, 5G penetration, subscriber growth — acts as a necessary enabler of digital economic growth but accounts for little independent predictive variance once literacy and economic development are controlled for. Literacy consistently dominated feature importance rankings, Elastic Net regularisation automatically discarded all telecom variables, and the interaction model confirmed theoretical complementarity between connectivity and human capital.

This framework should be understood as an associative rather than causal claim. The cross-sectional design of this study identifies correlational structure across Indian states at a single point in time; establishing the causal mechanism by which literacy translates into digital adoption requires panel identification or natural experiment designs beyond the scope of this analysis.

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